Physics-Informed Neural Networks for 2D Elasticity: Application to a Cantilever Beam Benchmark

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November 2025

Technical summary derived from a detailed project report

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1 Introduction

Finite element methods (FEM) are widely used to predict displacement and stress fields in mechanical structures, but their computational cost can become prohibitive in optimization, parametric studies, or real-time applications. These limitations have motivated the development of surrogate models capable of approximating mechanical responses at a reduced computational cost.

Within the field of Scientific Machine Learning, physics-informed approaches aim to combine data-driven models with governing physical laws in order to improve generalization and reduce the reliance on large training datasets. Physics-Informed Neural Networks (PINNs) belong to this class of methods by embedding the governing equations and boundary conditions directly into the training loss, enabling mesh-free approximation of physical fields.

In this work, a PINN is applied to a two-dimensional linear elasticity problem corresponding to a cantilever beam under transverse loading. The objective is to evaluate the ability of the model to reconstruct displacement and stress fields and to assess training behavior and accuracy. A finite element solution is used as a reference in order to quantify the performance of the physics-informed surrogate on a controlled benchmark problem.

2 Problem Definition

We consider a two-dimensional linear elastic cantilever beam subjected to transverse loading and modeled under a plane stress assumption. The problem is static and body forces are neglected. This benchmark configuration is commonly used to assess numerical methods in solid mechanics and provides a controlled setting to evaluate physics-informed surrogate models.

Geometry and Material Properties

The computational domain $\Omega \subset \mathbb{R}^2$ corresponds to the mid-plane of a rectangular beam of length $L = 500$ mm and height $h = 50$ mm, with coordinates

$$x \in [0, L], \qquad y \in [0, h].$$

The out-of-plane thickness is constant and equal to $t = 2.0$ mm. The material is assumed homogeneous, isotropic, and linearly elastic, with Young's modulus $E = 55{,}000$ MPa and Poisson's ratio $\nu = 0.26$. A plane stress formulation is adopted, which is appropriate for slender beams and thin plates.

Governing Equations

Let $\mathbf{u} = (u, v)^\top$ denote the in-plane displacement field. The equilibrium equations of linear elasticity in the absence of body forces read

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0} \quad \text{in } \Omega.$$

Under the small-strain assumption, strains are defined as

$$\varepsilon_{xx} = \frac{\partial u}{\partial x}, \quad \varepsilon_{yy} = \frac{\partial v}{\partial y}, \quad \gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}.$$

For an isotropic material under plane stress, stresses are related to strains through Hooke's law. The resulting system of partial differential equations constitutes the physics constraints enforced within the PINN formulation.

Boundary Conditions

The left boundary ($x = 0$) is fully clamped,

$$\mathbf{u} = \mathbf{0},$$

while the top and bottom edges are traction-free. A vertical resultant force $F_{\text{tot}} = 120$ N is applied at the free end ($x = L$). In the PINN framework, this loading is enforced through an integral constraint on the shear stress along the right boundary, ensuring global force equilibrium.

Reference Solution

For validation purposes, a finite element solution is used as the primary reference. In addition, classical Euler–Bernoulli beam theory provides analytical expressions for deflection and bending stress, which are employed as secondary reference quantities to assess the physical consistency of the predicted fields. These reference solutions enable a quantitative comparison between the PINN predictions and established mechanical models.

3 Methodology: PINN Formulation

Neural representation and automatic differentiation

A physics-informed neural network (PINN) is used as a smooth surrogate model of the displacement field. A fully-connected multilayer perceptron (MLP) maps spatial coordinates to the in-plane displacement

$$(x, y) \;\mapsto\; \mathbf{u}_\theta(x, y) = \big(u_\theta(x, y),\, v_\theta(x, y)\big)^\top,$$

where θ denotes the trainable parameters. To improve numerical conditioning, the input coordinates are linearly normalized to $[-1, 1]^2$. All spatial derivatives required by linear elasticity are computed via automatic differentiation, enabling the direct evaluation of strains, stresses, and equilibrium residuals in the strong form.

Physics constraints and loss function

The PINN is trained by minimizing a weighted sum of physics-based terms,

$$\mathcal{L}(\theta) = \sum_k w_k \, \mathcal{L}_k(\theta),$$

grouped as follows:

- **Interior physics:** least-squares enforcement of equilibrium ($\nabla\cdot\boldsymbol{\sigma} = \mathbf{0}$) at interior collocation points.
- **Boundary conditions:** clamped left edge (also enforced *hard* by construction), and traction-free top/bottom edges via stress penalties.
- **Global loading:** an end-load resultant constraint on the right boundary,

$$\int_{\Gamma_R} \sigma_{xy}(L, y)\, t \, \mathrm{d}y \approx -F_{\mathrm{tot}}.$$

- **Beam-theory priors (stabilization):** neutral axis (zero axial resultant), bending moment balance, curvature prior $\partial^2 v_\theta/\partial x^2 \approx M(x)/(EI)$, plus tip-deflection and right-edge uniformity constraints to reduce spurious end effects.

Sampling and optimization

Training uses a mixture of structured mesh-based points (reusing the FEM mesh nodes) and randomly sampled interior collocation points to improve domain coverage. Boundary point sets are extracted from the mesh for the clamped, traction-free, and loaded edges. The network is optimized with Adam (initial learning rate 10^{-3}) with a learning-rate scheduler, and the best checkpoint (lowest total loss) is retained for post-processing. Since the objective combines heterogeneous terms, a gradient-conflict mitigation strategy (PCGrad) is employed to reduce destructive interference between loss components and improve training stability.

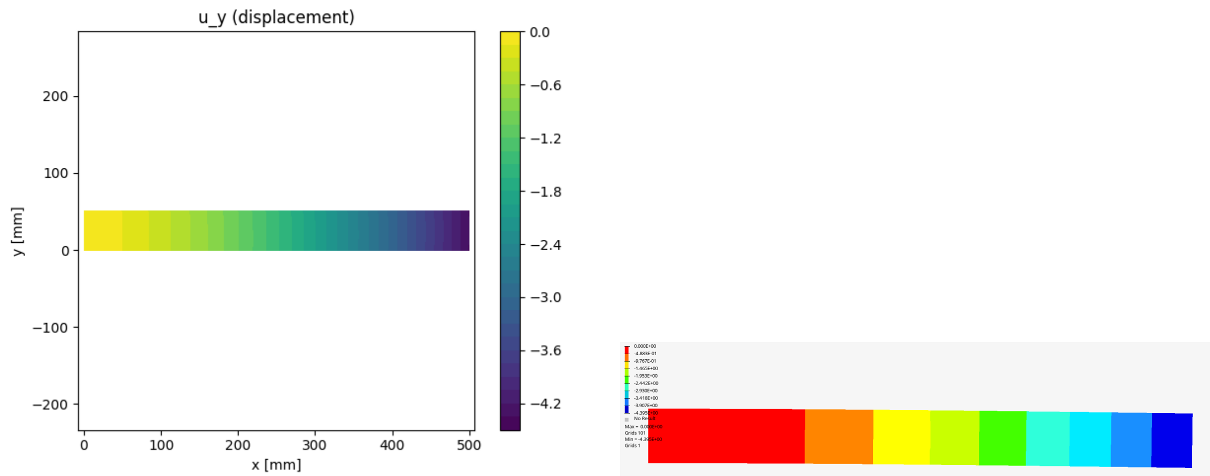
4 Results

This section summarizes the displacement and stress fields predicted by the PINN and compares them qualitatively to the finite element (FE) reference solution. Beam-level quantities and training convergence are also reported. Detailed error norms against FE are deferred to a dedicated benchmark section.

Displacement field

Figure 1 compares the vertical displacement field $u_y(x, y)$ predicted by the PINN to the FE reference. Both methods recover the expected bending-dominated response: u_y varies primarily along the beam axis with limited thickness-wise variation, and the maximum downward deflection occurs at the free end.

At $x = L$, the PINN predicts a mean right-edge deflection $v_{\text{right}}^{\text{PINN}} \approx -4.3616$ mm, while Euler–Bernoulli theory gives $v_{\text{EB}} \approx -4.3643$ mm and the FE solution yields $v_{\text{right}}^{\text{FE}} \approx -4.395$ mm. The corresponding relative deviations with respect to v_{EB} are 0.06% (PINN) and 0.70% (FE). Since the tip deflection is explicitly constrained in the loss, agreement with Euler–Bernoulli compliance is expected; the main discriminant therefore lies in the stress field and local equilibrium.



(a) PINN

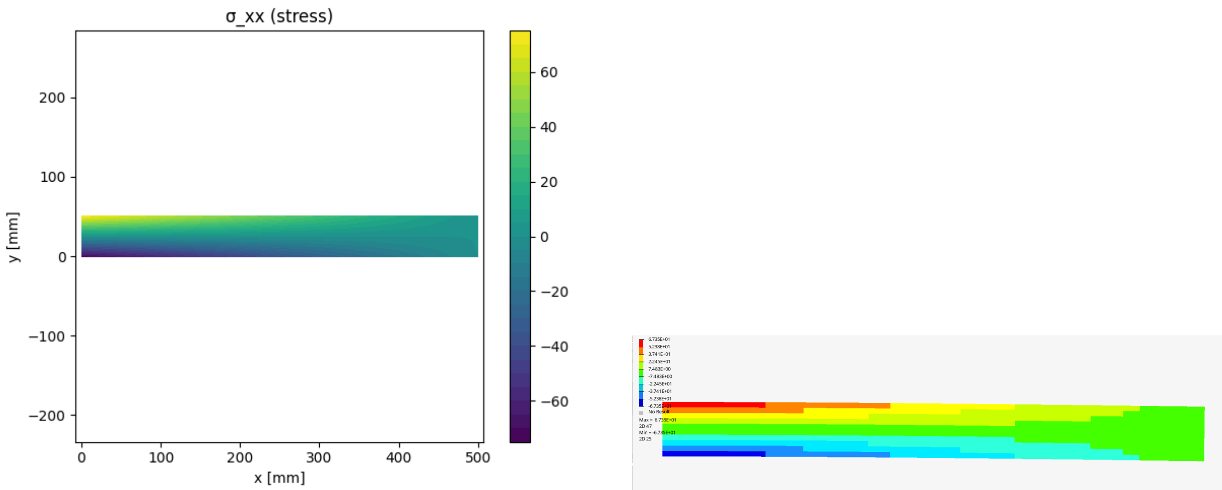
(b) FE

Figure 1: Vertical displacement field $u_y(x, y)$ (mm): comparison between the PINN prediction and the FE reference solution.

Stress field

Figure 2 shows the normal stress component $\sigma_{xx}(x, y)$ recovered from the predicted displacement field using the plane-stress constitutive law. Both PINN and FE solutions exhibit the expected nearly linear distribution of σ_{xx} across the thickness and peak stresses near the clamped edge. The PINN contours remain smooth, with minor deviations localized near the fixed boundary.

Using Euler–Bernoulli beam theory as a secondary reference, the theoretical peak bending stress magnitude at the clamped outer fibre is $|\sigma_{xx,\max}^{\text{EB}}| \approx 72.012$ MPa. The PINN predicts $|\sigma_{xx,\max}^{\text{PINN}}| \approx 71.609$ MPa (relative deviation 0.6%), while the FE model gives $|\sigma_{xx,\max}^{\text{FE}}| \approx 67.35$ MPa (relative deviation 6.5%), consistent with an underestimation due to mesh resolution in the present FE setup.



(a) PINN

(b) FE

Figure 2: Normal stress field $\sigma_{xx}(x, y)$ (MPa): comparison between the PINN prediction and the FE reference solution.

Training convergence

Figure 3 reports the convergence of the total loss and selected components (log scale). The PDE residual decreases rapidly and then reaches a plateau, indicating stable interior equilibrium enforcement. Boundary traction losses show a similar stabilization, while beam-level constraints converge more slowly. The tip deflection term exhibits a sharp drop when it becomes dominant in the optimization, consistent with the accurate right-edge compliance reported above.

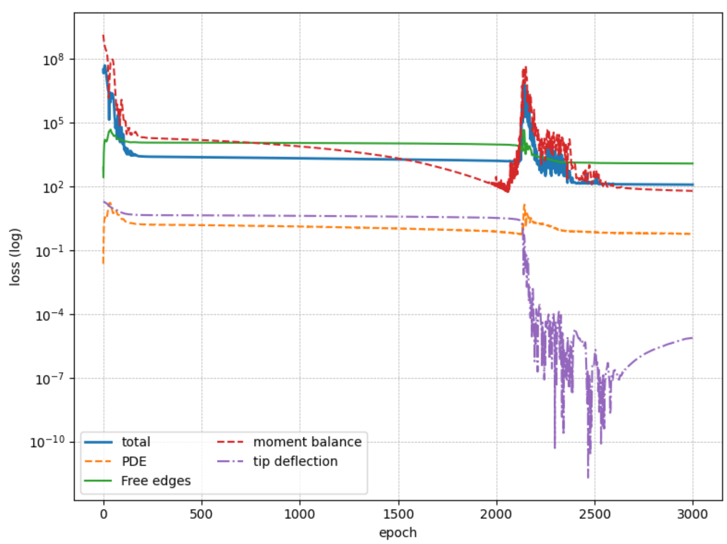


Figure 3: Convergence history of the total loss and selected components during training (log scale): PDE residual, free-edge boundary term, moment balance, and tip-deflection constraint.

Summary of key quantities

Table 1 compares tip deflection and maximum bending stress magnitudes predicted by Euler–Bernoulli theory, the PINN, and the FE reference.

Quantity	Theory (EB)	PINN	FE
Tip deflection v_{right} [mm]	-4.3643	-4.3616	-4.395
Max bending stress $\lvert\sigma_{xx,\max}\rvert$ [MPa]	72.012	71.609	67.35

Table 1: Comparison of beam-level quantities for Euler–Bernoulli theory (EB), the PINN, and the FE reference solution. For stresses, absolute values are reported.

5 Benchmarking Against FEA

Finite element reference

A linear static plane-stress finite element analysis is used as a numerical reference. The model is solved in OptiStruct using four-node quadrilateral elements (CQUAD4), with a structured mesh of 100×10 elements (typical element size ≈ 5 mm in both directions). The left edge ($x = 0$) is fully clamped ($u_x = u_y = 0$), and a downward tip load $F_{\text{tot}} = 120$ N is applied at $x = L$. Nodal displacements and stresses are exported and used to evaluate the PINN at the same node locations.

Error metrics

To quantify agreement, the PINN predictions are evaluated at the FEA nodes and compared field-wise. For each scalar field $f \in \{u_x, u_y, \sigma_{xx}\}$, relative discrete L^2 and L^∞ errors are computed:

$$\varepsilon_{L^2}(f) = \frac{\left(\sum_{i=1}^{N} |f_i^{\text{PINN}} - f_i^{\text{FE}}|^2\right)^{1/2}}{\left(\sum_{i=1}^{N} |f_i^{\text{FE}}|^2\right)^{1/2}}, \qquad \varepsilon_{L^\infty}(f) = \frac{\max_i |f_i^{\text{PINN}} - f_i^{\text{FE}}|}{\max_i |f_i^{\text{FE}}|}.$$

Field-level comparison

Figure 4 illustrates the pointwise discrepancy for σ_{xx}. Errors are mainly concentrated near the clamped boundary, where stress gradients are largest and strong-form PINNs tend to smooth sharp variations. Away from the support, discrepancies remain comparatively small.

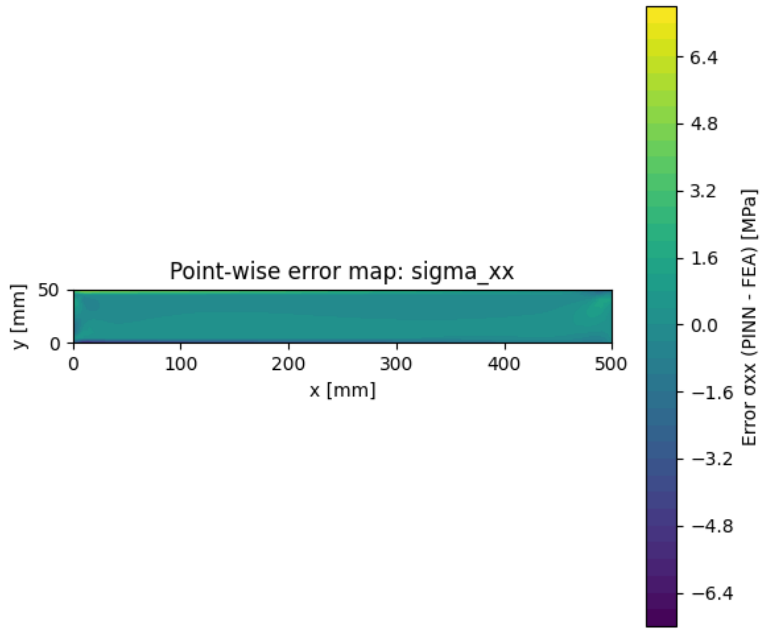


Figure 4: Pointwise error map for axial stress: $e_{\sigma_{xx}}(x, y) = \sigma_{xx}^{\mathrm{PINN}}(x, y) - \sigma_{xx}^{\mathrm{FE}}(x, y)$ (MPa), evaluated at the FEA nodes.

Quantity	ε_{L^2}	ε_{L^∞}
u_x	5.08×10^{-2}	6.94×10^{-2}
u_y	4.08×10^{-2}	3.43×10^{-2}
σ_{xx}	6.84×10^{-2}	1.10×10^{-1}

Table 2: Relative L^2 and L^∞ errors between the PINN prediction and the FEA reference solution.

6 Data-Assisted PINN Fine-Tuning

A second training stage investigates whether incorporating a limited amount of FEA data can reduce field-level discrepancies, especially on stresses. Starting from the best physics-only parameters, the PINN is fine-tuned using a hybrid physics-and-data loss:

$$\mathcal{L}_{\mathrm{hybrid}} = w_{\mathrm{PDE}}\mathcal{L}_{\mathrm{PDE}} + w_{\mathrm{BC}}\mathcal{L}_{\mathrm{BC}} + w_{\mathrm{tip}}\mathcal{L}_{\mathrm{tip}} + w_{\mathrm{data}}\mathcal{L}_{\mathrm{data}},$$

where $\mathcal{L}_{\mathrm{data}}$ penalizes mismatches between the PINN predictions and FEA values at mesh nodes for (u_x, u_y, σ_{xx}).

Table 3 compares physics-only and data-assisted errors. Displacement errors remain in the same range, confirming that physics-only training already captures kinematics well. In contrast, the agreement on σ_{xx} improves noticeably: both the global L^2 error and the worst-case L^∞ error decrease, showing that modest supervision can effectively correct local stress discrepancies without compromising global mechanical consistency.

Quantity	Metric	Physics-only PINN	Data-assisted PINN
u_x	ε_{L^2}	5.08×10^{-2}	6.49×10^{-2}
u_y	ε_{L^2}	4.08×10^{-2}	4.25×10^{-2}
σ_{xx}	ε_{L^2}	6.84×10^{-2}	5.27×10^{-2}
u_x	ε_{L^∞}	6.94×10^{-2}	8.13×10^{-2}
u_y	ε_{L^∞}	3.43×10^{-2}	3.55×10^{-2}
σ_{xx}	ε_{L^∞}	1.10×10^{-1}	6.72×10^{-2}

Table 3: Relative L^2 and L^∞ errors vs. FEA for the physics-only and data-assisted PINN models.

7 Discussion

Why the physics-only PINN works

The physics-only PINN reproduces the cantilever response with good accuracy at both beam level and field level (Sections 4–5). This performance is largely explained by three factors.

First, the benchmark is linear, well-posed, and dominated by smooth bending kinematics. In such a regime, equilibrium and boundary conditions provide strong constraints on the solution structure.

Second, the training objective combines complementary information across scales: local enforcement (PDE residual and boundary tractions) and global anchoring through beam-inspired constraints (moment balance and tip compliance). This multi-term design stabilizes training and reduces common failure modes in elasticity PINNs. The resulting fields are consistent with the qualitative patterns observed in Figures 1 and 2.

Third, the network capacity is sufficient to represent the smooth displacement field, and automatic differentiation provides consistent derivatives for strains and stresses. Once a reasonable loss-weight balance is achieved, optimization converges to a physically consistent compromise, as reflected by the loss history in Figure 3.

Effect of data-assisted fine-tuning

Fine-tuning with a hybrid physics-and-data loss primarily improves local stress accuracy. While displacement errors remain in a similar range, the axial stress agreement improves: the relative L^2 error on σ_{xx} decreases from 6.84% to 5.27%, and the maximum relative error drops from 11.0% to 6.72% (Table 3). This indicates that limited supervision is particularly beneficial for stress fields, which depend on spatial derivatives of the learned displacement and are therefore more sensitive to small kinematic inaccuracies.

Strong-form limitations near high-gradient regions

The main discrepancies occur near the clamped boundary, where stress gradients are largest (Figure 4). This is consistent with known limitations of strong-form PINNs: (i) smooth neural representations tend to under-resolve boundary layers unless sampling is adapted, and (ii) stress computation amplifies small displacement errors through derivatives. In addition, competing loss terms (local equilibrium vs. global constraints) can lead to suboptimal compromises in gradient-dominated zones.

Role relative to classical FEA

For a single geometry and single load case, classical FEA remains the most efficient approach: it provides a deterministic solution with controllable discretization error at low cost. In contrast, the PINN requires an upfront training phase. However, once trained, it acts as a mesh-free, differentiable surrogate that can be evaluated at arbitrary points at negligible marginal cost. The main value of the approach therefore lies in regimes where repeated evaluations are needed (e.g. parametric studies, optimization loops, or digital-twin settings), and where differentiability can be exploited for gradient-based workflows.

8 Conclusion

This work evaluated a physics-informed neural network for a 2D plane-stress cantilever beam and benchmarked it against a finite element reference. The physics-only PINN accurately recovered global compliance and produced displacement and stress fields that agree with FEA within a few percent in relative norms. A data-assisted fine-tuning stage further reduced stress discrepancies, notably decreasing the worst-case error on σ_{xx} while preserving displacement accuracy.

The study highlights both the promise and limitations of strong-form PINNs in elasticity: they can generate mechanically consistent surrogates on smooth benchmarks, but stress accuracy remains sensitive to boundary layers, sampling, and loss balancing. Future work should focus on (i) parameterized surrogates that generalize across loads/materials, (ii) adaptive sampling or weak/energy-based formulations to better capture high-gradient regions, and (iii) mesh-based architectures such as graph neural networks for complex geometries.